

DATE: ___/___/___

DAY: ___/___/___

Name:-

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SUBJECT:-

DATA STRUCTURE.

ASSIGNMENT : 02

date: 23.2.26.

Question : 01

A singly linked list contains nodes where each node has 2 parts. Data and Next (pointer to the next node)

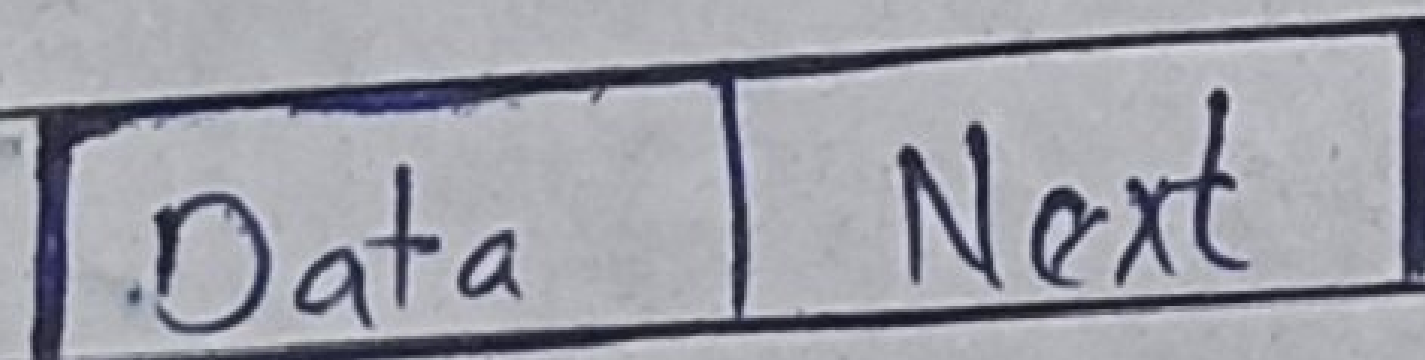
Explain:-

The following element of a singly linked list:-

↳ Node:-

The basic building block of a linked list.

Each node contains two parts: Data and Next.



↳ Data Fields:-

Stores the value of the node

e.g. 10, 20, 30.

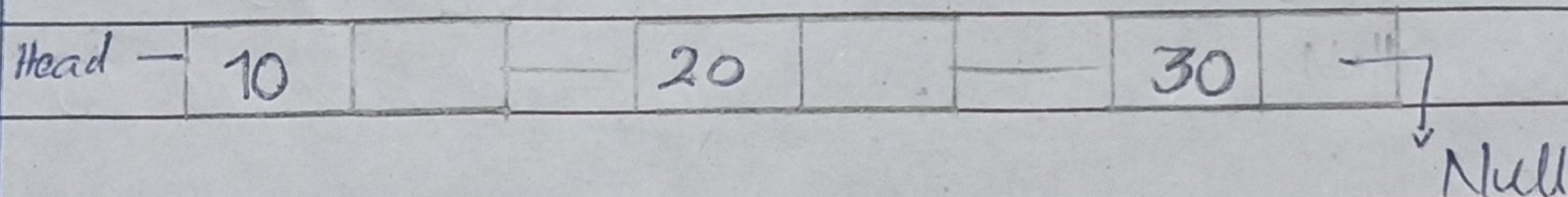
↳ Next Pointers:-

Stores the address of the next node. If Next = Null, it indicates the last node.

↳ Head Pointers:-

Point to the first node in the list. If the list is empty, Head = Null.

Example 8-



(Part : a)

(Insertion Operation)

Write pseudocode and Dry run for

1. Insertion at Beginning:-

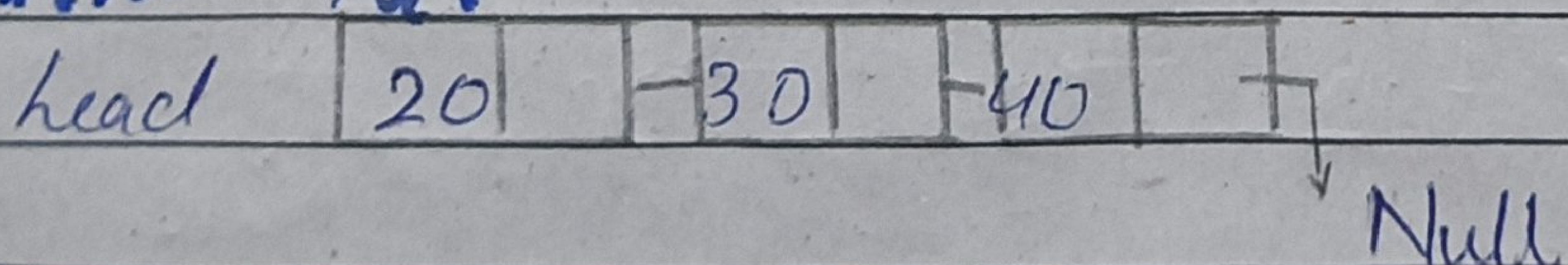
Pseudo Code:-

```
newnode = create node
newnode.data = Value
newnode.next = head
head = newnode
```

Return head

Insert 10 into list: 20 → 30 → 40

initial list:



Dry run:-

Action

Pointer Movement

Create new node

new node

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Set newnode.data = 10
newnode.next = head
head = newnode

newnode 20

head now at 10

After Insertions- Final list.

Head → 10 → 20 → 30 → 40 → Null

2. Insertion at End:-

Pseudo Code:-

```
newnode = create node
newnode.data = value
newnode.next = Null
if head == Null
    head = newnode
    Return head
ptr = head
while ptr.next != Null:
    ptr = ptr.next
ptr.next = newnode
Return head
```

Insert 50 into list: 10 → 20 → 30

Initial list:

head → 10 → 20 → 30 → Null

Dry run

Create new node 50

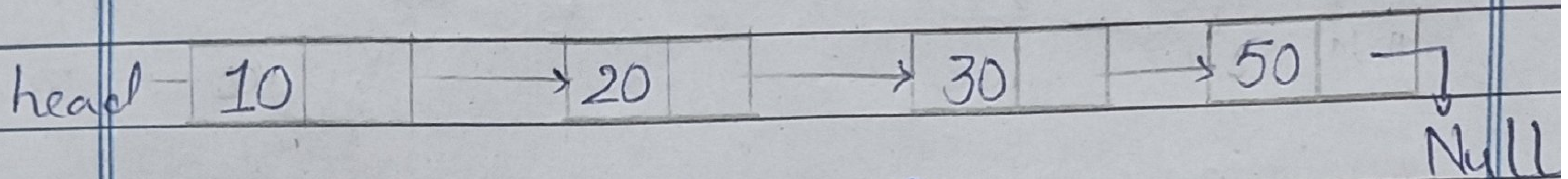
ptr starts at head 10

Move ptr to 20

Move ptr to 30

ptr.next = newnode

Final list:-



3. Insertion at Specific Position:

insert 25 at position 3 in list 10, 20, 30, 40

Pseudo Code:-

newnode.next = head

head = newnode

Return head

ptr = head

i = 1

while i < pos - 1:

ptr = ptr.next

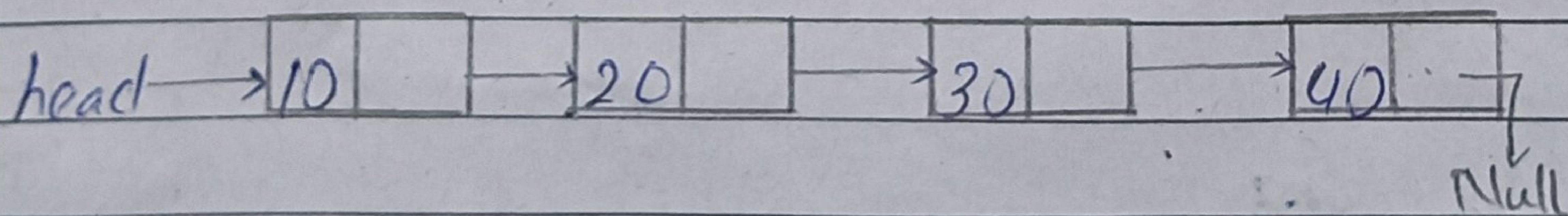
i = i + 1

newnode.next = ptr.next

ptr.next = newnode

Return head

Initial list:-



Day wise - Action.

Create new node 25

Pointer.

new node

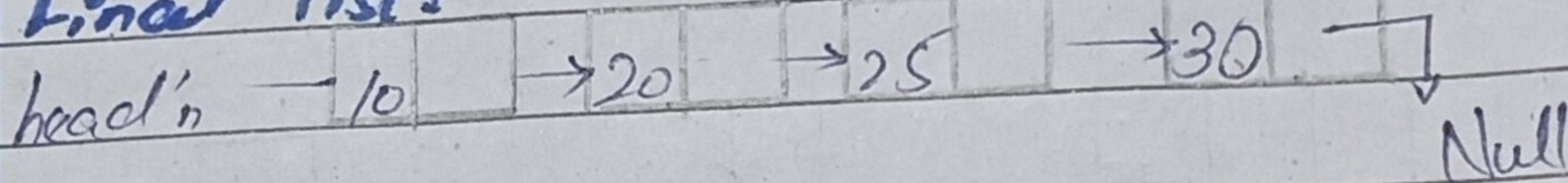
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Move ptr to pos -1 = 2
 new node . next = ptr . next
 ptr . next = new node

ptr 'n 20
 25 'n 30
 20 'n 25

Final list:



(Part - B) (deletion operations)

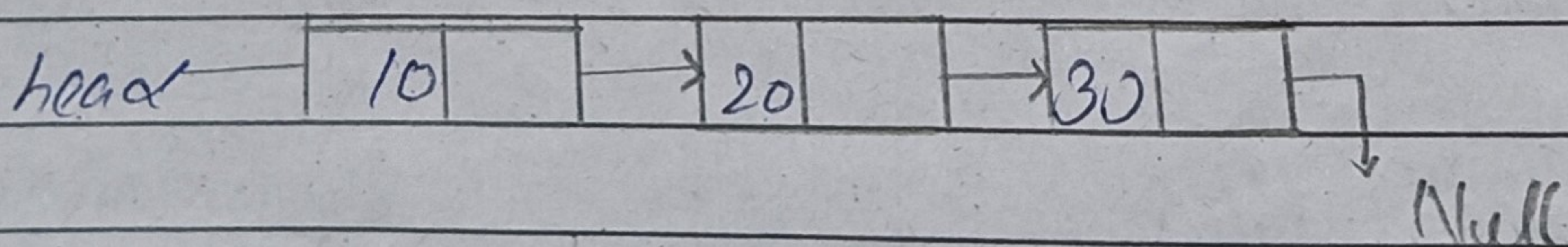
4. **Deletion from Beginning:-**

delete first node from list : 10 20 30

Pseudo Codes:-

ptr = head
 head = head . next
 delete ptr
 Return head

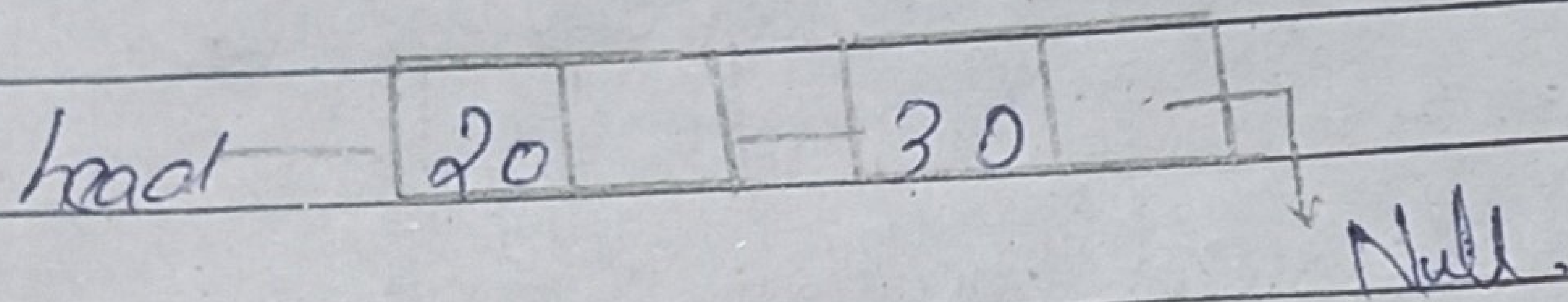
Initial list:-



Dry runs:-

ptr = head 10
 head = head . next 20
 delete ptr

Final list :-



5. Deletion from End:-

delete last node from list : 10 20 30 40

ptr = head

prev = Null

while ptr . next != Null:

prev = ptr

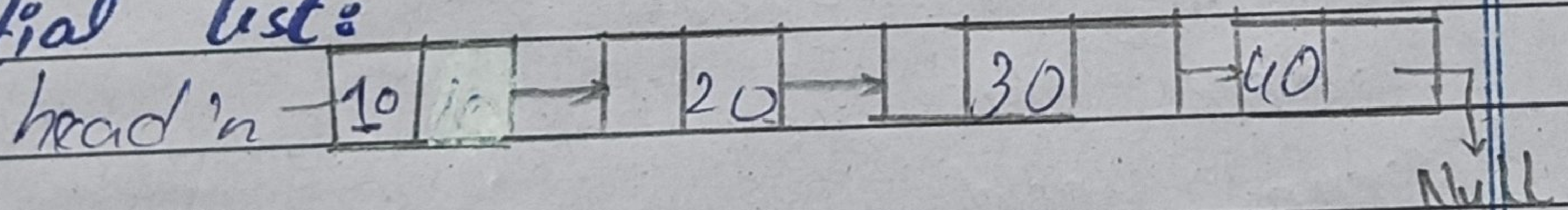
ptr = ptr . next

prev . next = Null

delete ptr

Return head.

Initial list:



Day Run:-

ptr = 10

ptr = 20, prev = 10

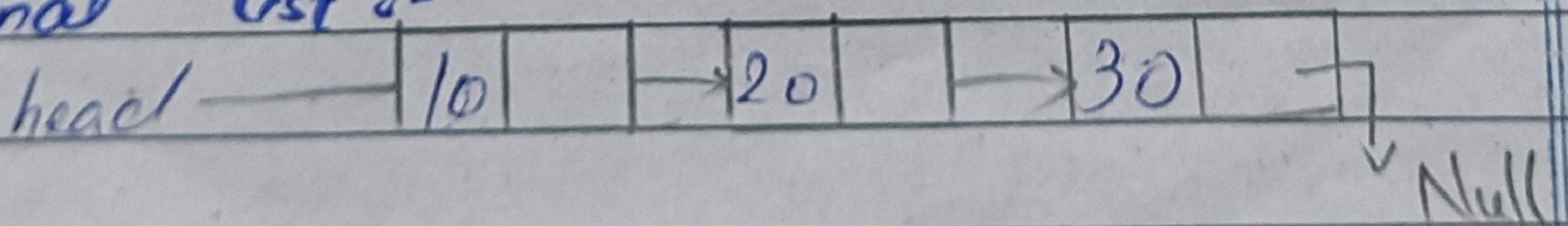
ptr = 30, prev = 20

ptr = 40, prev = 30

prev . next = Null

delete ptr

Final list:-



6. Deletion from a Specific Position.

deletion node at position 2 from
10ⁿ 20ⁿ 30ⁿ 40ⁿ

ptr = head

prev = Null

i = 1

if pos == 1

head = head.next

delete ptr

Return head

While i < pos:

prev = ptr

ptr = ptr.next

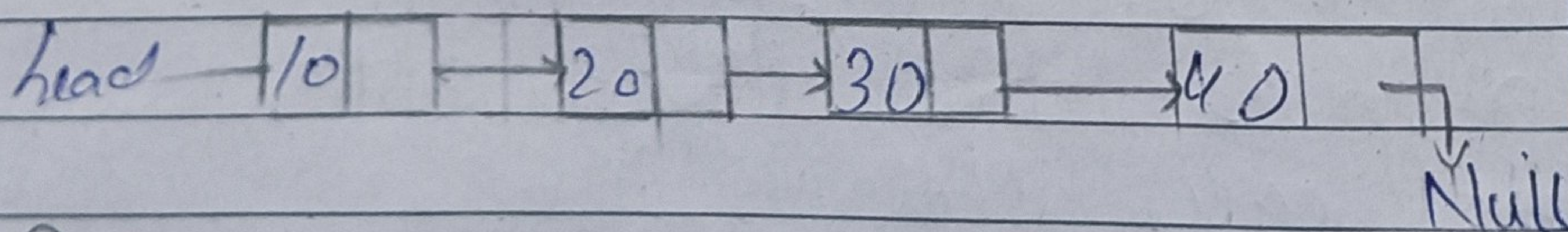
i++

prev.next = ptr.next

delete ptr

Return head

Dry Run -
Initial list



Delete position 2ⁿ node = 20.

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ptr'n 10, prev = Null

ptr'n 20, prev'n 10

prev.next = ptr.next 10'n 30

delete ptr

Final list:-

